Securities Trading: Principles and Procedures, Chapter 11

Private information

If only one person knows something, how can it be reflected in the security price?

Readings
- STPP Chapter 11
- Bagehot, W., 1971. The Only Game in Town. Financial Analysts Journal 27, no. 2 (March-April), 12-22, (pseudonym for Jack Treynor).

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Efficiency and private information: an example

- ABC stock is offered at $51.
- Ivan (*only* Ivan) knows: there will a takeover bid for ABC at $60.
  - Ivan lifts ABC's offers, buying until the offer reaches $60.
  - *Note: this is illegal in the US*
- Other traders see that ABC is offered at $60.
  - They don't know why (until the takeover bid is announced).
- The bid is made public, Ivan sells his stock.

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For this to work…

- Ivan has to trade.
- Ivan's information (the takeover bid) must become public.
- Ivan is a short-term information trader.
- A long-term investor does not need the information to become public.
  - She can buy, hold, and collect higher dividends.

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Problems with the story

- Before the takeover bid is announced, there's no new fundamental public information.
  - Once Ivan lifts the offers at $51, why don't sellers continue to offer at $51?
  - If someone is willing to pay above the $51 ("public information") value, why don't short sellers step in?
  - How can the offer ever reach $60 before the revelation of the takeover bid?

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Using limit orders in the F1 game

- Suppose at the start of the game, I enter
  - "Buy limit $1,049" (a bid)
  - "Sell limit $1,051" (an ask/offer)"
- If the first news announcement is good, my offer will be lifted.
  - I've sold at $1,051, so I'm short.
  - … and the price is rising.
- If the first news announcement is bad, my bid will be hit.
  - I've bought at $1,049. I'm long, and price is falling.
- Limit orders tend to put us on the wrong side of large price movements.

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Using limit orders in the MM game

- No major price changes, nobody is trading based on information.
  - When our limit orders execute, we don't find ourselves on the wrong side of the price movement.
- The computerized market makers are not aggressive and competitive.
  - They allow us to make bids and offers with a wide spread.
  - The wide spread generates profits.

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The market maker

- A market maker posts bid and ask quotes.
- There are two kinds of market conditions.
  - If the world looks like MM, she makes money.
  - … F1, she loses money.
- Alternatively: there are two kinds of traders.
  - If the traders are *uninformed* (they don't know what's going to happen), she makes money.
  - If the traders are *informed* (they know what's going to happen), she loses money.
  - Her profits on the uninformed traders must offset the losses to the informed traders.

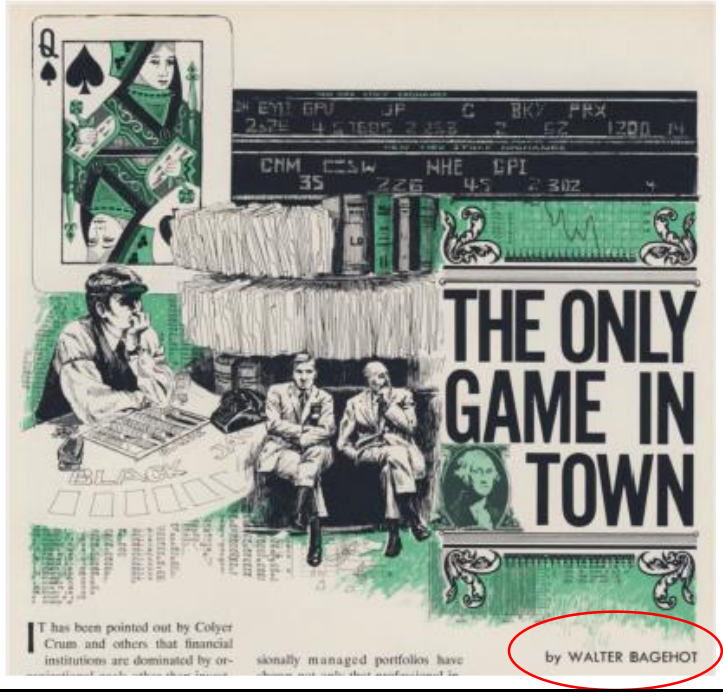
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We'll show that more private information …

- Increases the *bid-ask spread*.
  - Informed traders impose costs on dealers (and everyone else)
- Causes buy and sell orders to have long-term impacts on security prices (*order-price impact*)
  - Buy and sell orders are signals because they might have originated from informed traders.
- *Market failure*: in extreme cases, the market will shut down

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- "Walter Bagehot"
  - The real Walter Bagehot was a 19th c. British journalist.
  - Our "Walter Bagehot" is a pen name used by Jack Treynor (Treynor Capital Management)
- The article appeared in The Financial Analysts Journal (March/April 1971)


THE ONLY GAME IN TOWN

by WALTER BAGEHOT

IT has been pointed out by Colyer Crum and others that financial institutions are dominated by or-

sionally managed portfolios have

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Treynor's/Bagehot's logic

- Many investors lose money in the market because they can't overcome trading costs.
- A significant portion of the trading cost comes from the bid-ask spread.
- Bids and asks are posted by the market-maker (dealer or specialist)
  - Or, nowadays, a limit order trader

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The dealer faces two types of traders

- *Liquidity traders* (*uninformed traders*)
  - They buy because they have extra cash; they sell to raise cash.
  - Their needs are unrelated to the stock's value.
  - Buyers and sellers arrive at equal rates.
  - The dealer earns the half-spread on each arrival.
- *Informed traders* know in advance things that change the stock's value.
  - Informed order flow is one-sided.
  - They buy on positive news; sell on negative.
  - The dealer always loses to these traders.

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Informed trading and the bid-ask spread

- The dealer's expected losses to informed traders depend on
  - How many are there?
  - How much do they trade?
  - How accurate is their information.
- To stay in business, the dealer's profits from the liquidity traders must exceed the losses to informed traders.
- To maintain profits, when there are more informed, the dealer must offset the increased losses by increasing profits from the liquidity traders.
  - The dealer raises the bid-ask spread.

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More on the bid-ask spread

- So far, we've viewed MM (dealer) profits as a reward for patiently maintaining a market presence.
- Treynor:
  - MMs also incur losses to informed traders.
  - They have to set a wider bid-ask spread.
  - This is costly to uninformed traders.
  - MM's losses to informed traders are passed on to uninformed traders.

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A simple *asymmetric information* model

- The end-of-day value of a share of stock, V, is random.
- At market close: either $V = High = 150$ or $V = Low = 100$
  - With equal probability.
- Before trading starts, the expected value is
  - $EV = \frac{1}{2}100 + \frac{1}{2}150 = 125$
  - This is the unconditional expected value.
- There are two types of customers
  - Informed traders know V.
  - Uninformed ("retail") traders don't know V.
- The dealer doesn't know V

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Traders' behavior

- Uninformed traders are equally likely to buy or sell.
- Informed traders
  - If they know that $V = High$, they *always* buy.
  - If they know that $V = Low$, they *always* sell.
- After the dealer sets the bid and offer quote, a trader arrives. The trader's type is random.
  - Suppose $P(informed) = 0.2$.
    - That is, 20% of the incoming traders are informed.
  - $P(uninformed) = 0.8$.

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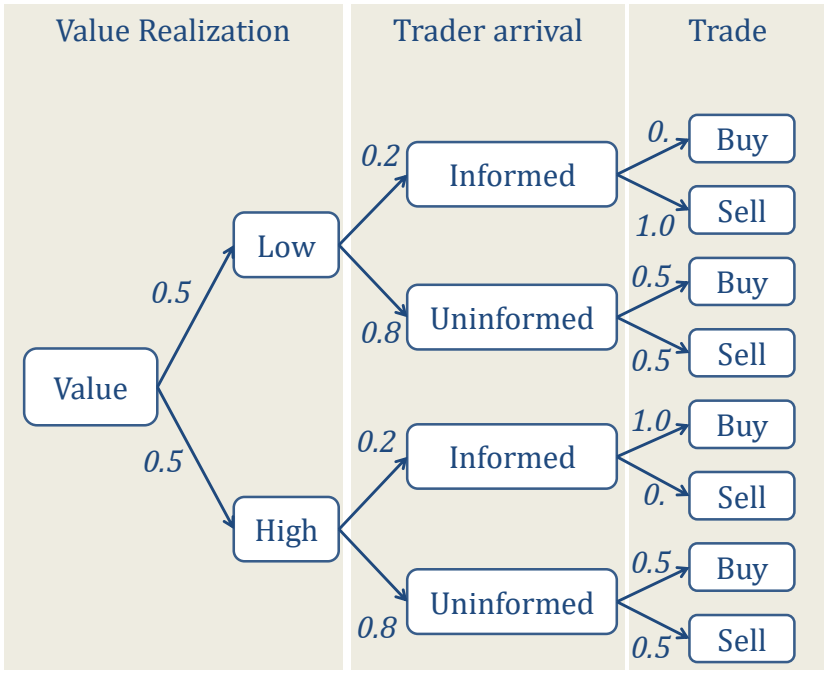
The first quotes and the first trade

- The dealer sets bid and ask quotes.
- A random trader arrives and buys or sells.
  - The dealer doesn't know if she is informed or not.
- How should the dealer set their bids and asks?
- Next slides show the *event tree*, the sequencing of possible events
  - and *conditional* probabilities.
  - or *joint/total* probabilities

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Conditional probabilities are in italics

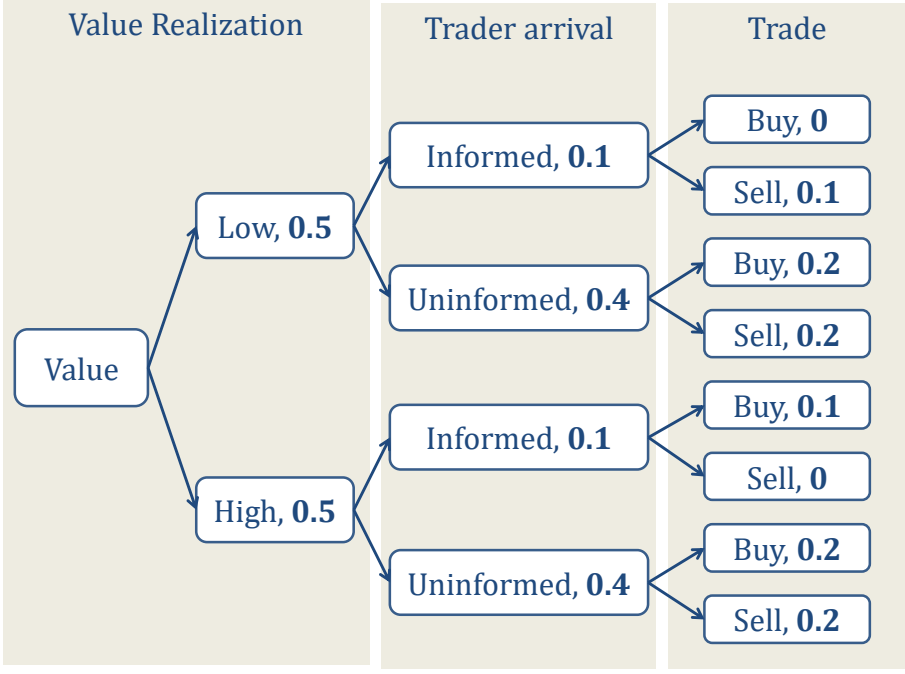
"Given that we're *here* the probability of *going there next* is …"



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Total probabilities in bold

Starting with the value draw, the probability that we'll end up *here* is …

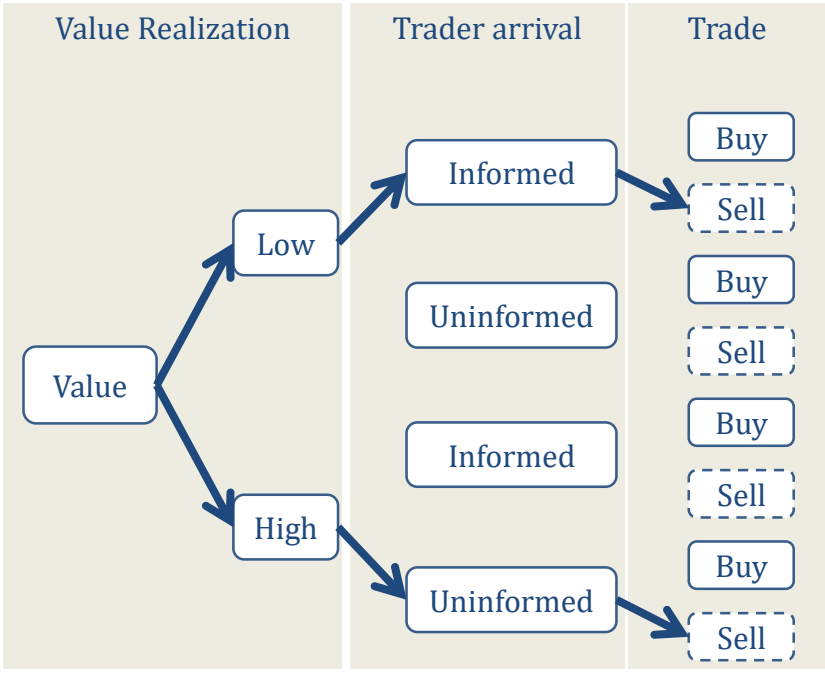


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After the trade, what does the dealer know?

- The dealer knows whether the trader has bought or sold.
- The total probability of a buy is 0.5
  - The total probability of sell is also 0.5
- The dealer doesn't know for sure what path ended up in "buy" or "sell".
- Suppose the incoming trader sells …

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Conditional probability

- Suppose that the bid is hit (the incoming order is *sell*).
- Before the *sell*, the dealer thinks that $P(V = Low) = 0.50$.
- After the *sell*, want to know the conditional probability $P(V = Low|sell)$
  - "The probability of a low value given a *sell*."
- $P(V = Low|Sell) = \frac{P(V=Low,\ Sell)}{P(Sell)} = \frac{0.1+0.2}{0.5} = 0.6$
- Note: in this particular case $P(V = Low|Sell) = P(Sell|V = Low)$
  - But in general, $P(A|B) \neq P(B|A)$
  - Example: in a draw of one card from a full deck of 52 cards: $P(Ace|Heart) = \frac{1}{13}$, but $P(Heart|Ace) = \frac{1}{4}$

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Conditional expectation

- The expected security value given *sell* is
  - $E[V|Sell]$
    $= P(V = Low|Sell) \times V_{Low} + P(V = High|Sell) \times V_{High}$
    $= 0.6 \times 100 + 0.4 \times 150 = 120$
  - Embedded problem 1: show that $E[V|Buy] = 130$.
    Answer is at the end of this handout.

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How should the dealer set his bid?

- If he bids 121 and he's hit, he'll have an (expected) loss.
- If he bids 119 and he's hit, he'll have an expected profit.
  - But someone else will compete, bidding, say, 119.10.
- The forces of competition will drive the bid to the point where $Bid = E[V|sell] = 120$
- What should be the ask?

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Suppose someone hits the dealer's bid (at $120)

- If seller is uninformed,
$$dealer's\ profit = \underbrace{\$125}_{EV} - \$120 = \$5$$
- If seller is informed,
$$dealer's\ profit = \underbrace{\$100}_{EV|informed} - \$120 = -\$20$$
- $$\underbrace{0.8}_{P(uninformed)} \times \$5 + \underbrace{0.2}_{P(informed)} \times (-\$20) = 0$$
- The dealer's expected profits from uninformed just cover the expected losses to the informed.

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Market failure

- As the spread widens, there are fewer uninformed traders.
  - "I'll buy/sell something else."
  - The revenue from the uninformed traders drops.
- The losses to the informed traders stay the same.
- At some point, the dealer (MM) is losing money.
  - She/he leaves the market.
  - The market fails.

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- Embedded problem 2:
  - Show that if the probability of an informed trader is 40% (rather than 20%) the market will be 115 bid, offered at 135.
  - Answer is at the end of this handout.

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Questions

- If an important news announcement is scheduled for 8:30, why might the spread widen *before* 8:30?
- A dealer might prefer to quote different spreads to different clienteles (retail, institutional, proprietary, etc.)
  - Can a dealer pay brokers to send her the retail customer orders?
- A dark pool can discriminate against customers more likely to be informed.
  - But an Exchange can't discriminate.
  - If we want to trade, where will we face the fewest informed traders?

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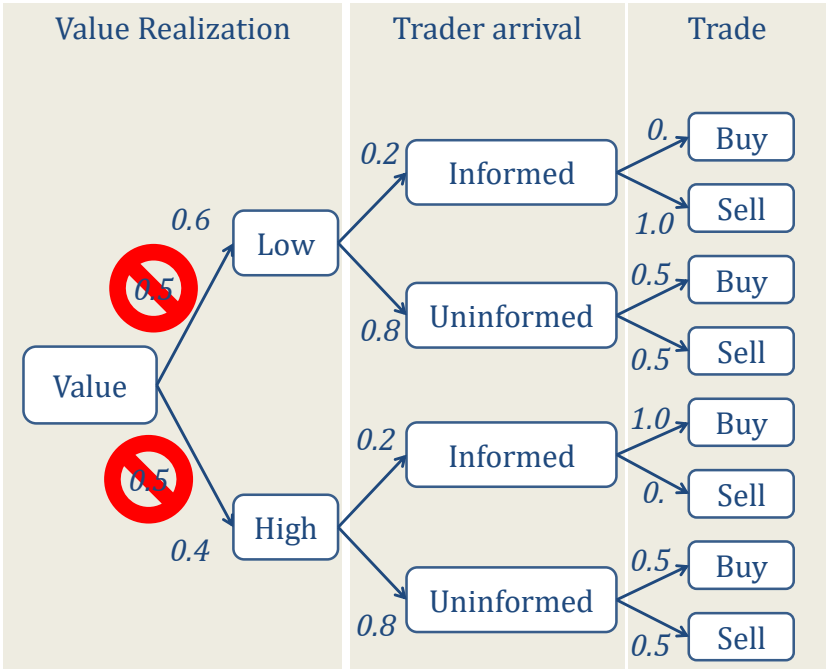
The first trader has just hit the dealer's bid. What's next?

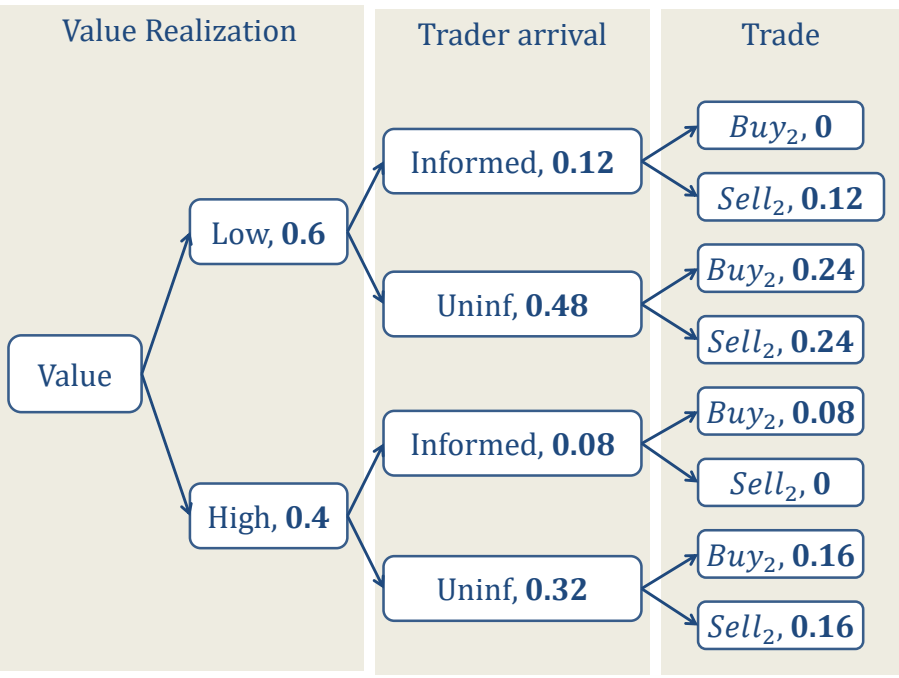
- Recap
  - $V = Low = \$100$ or $V = High = \$150$
  - $P(V = Low) = P(V = High) = 0.5$
  - For any incoming trader, $P(Informed) = 0.2$
  - The dealer sets $bid = \$120$ and $offer = \$130$
- Suppose that the incoming trader hits the bid (sells to the dealer).
- How should the dealer revise his quotes in anticipation of the second trade?

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- From the analysis leading up to the first trade, the dealer knows that
  $P(V = Low|Sell_1) = 0.6$ and $P(V = High|Sell_1) = 0.4$
  - where we've added a subscript: $Sell_1$ indicates that the first trade was a sell.
- Without knowing whether the first trade is a buy or a sell, the dealer thinks $P(V = Low) = 0.5$.
- The event tree leading up to the second trade has the same form as the first tree, but with this one probability changed.

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Analysis

- Conditional probabilities

$$P(V = Low|Sell_2) = \frac{P(V = Low,\ \ Sell_2)}{P(Sell)} = \frac{0.12 + 0.24}{0.12 + 0.24 + 0.16} = 0.692$$

$$P(V = High|Sell_2) = 0.308$$

- Conditional expectations

$$E(V|Sell_2) = P(V = Low|Sell_2) \times 100 + P(V = High|Sell_2) \times 150$$
$$= 115.385$$

- Buy the same logic as for the first trade, this should be the dealer's bid.
- Embedded problem 3: Show $P(V = Low|Buy_2) = 0.5$ and $E(V|Buy_2) = 125 \quad (= Ask)$. Answer at end of handout.

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Summary

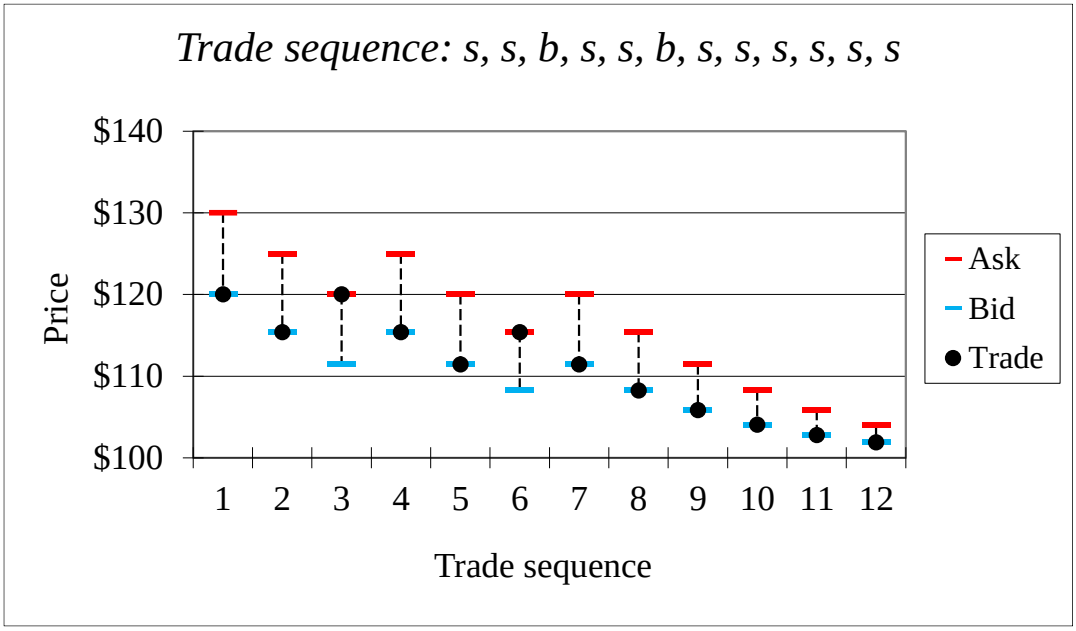
- Prior to *any* trade, $EV = 125$
- Prior to the first trade:
  $Bid = 120 = E[V|sell_1]$ and $Ask = 130 = E[V|buy_1]$
- Suppose the first trade is a *sell*. Everyone now thinks that the stock is worth 120.
- The new quotes prior to the second trade are:
  $Bid = E[V|sell_1, sell_2] = 115.385$ and
  $Ask = E[V|sell_1, buy_2] = 125$
- Order price impact: "The first *sell* drives down the price."

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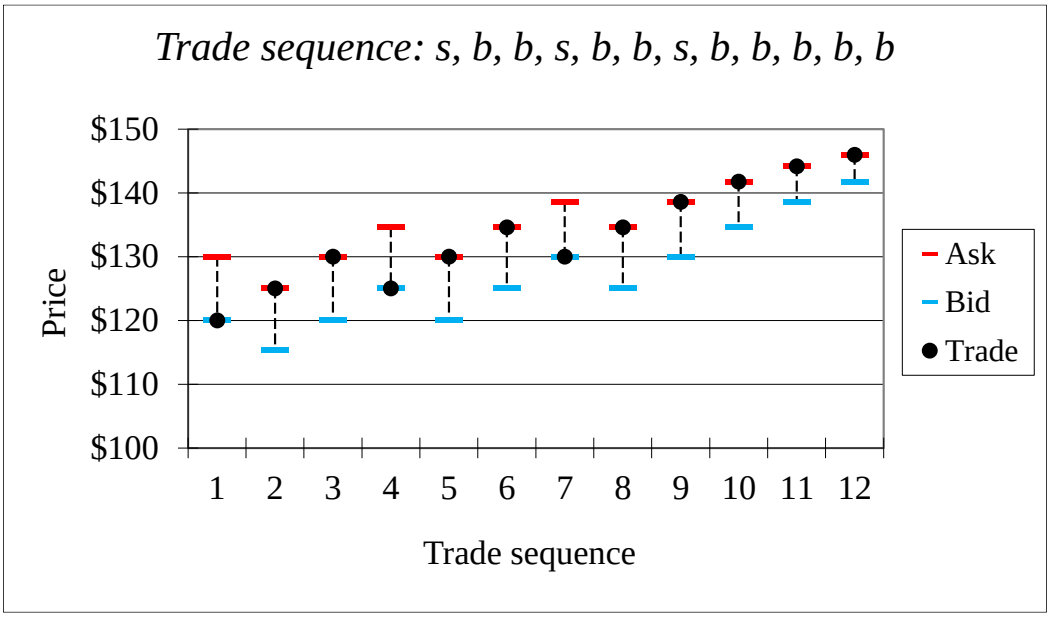
Trade sequences

- For any given hypothetical sequence buys and sells, we can construct the path of the bid and offer
- Generally, as trading advances:
  - The bid and offer move in the direction of the orders.
  - A *buy* drives the quotes up; a *sell* drives the quotes down.
- Since the informed tend to trade in one direction, the order flow tends to become increasingly one-sided,
  - The price converges toward the true value.
  - The spread drops (because there is less uncertainty).
- Next: some sample plots (see simplesequentialtrademodel.xlsx on NYU classes, > resources > spreadsheets)

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Order price impacts

- When a trade occurs, there *is* new public information: the trade itself (and, usually, its direction).
- The market interprets the order direction as a signal of private information, and adjusts the price accordingly.
- The market does not know which traders are informed.
  - Analysis of the trades, even in retrospect, won't tell us which traders (if any) were informed.
  - The market reaction tells us what the market *thought*, not the reality.

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Uninformed orders move the market just as much as informed orders

- The price movements in response to uninformed orders aren't warranted.
  - But they won't be corrected until there's authoritative information.
  - Or the sustained absence of new information.

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Trading and creation of *new* information asymmetries.

- Suppose I'm an uninformed trader, and I lift the offer.
- The bid and offer rise. The market (and market maker, if there is one) think, "Maybe he had some positive information."
- I know that I *don't* have any information. My knowledge of my own ignorance gives me an advantage over the market.
- I know (and only I know) that bids and offers are higher than they should be.
- Can an uninformed trader move the market so as to give the appearance of information?

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Open questions

- How long do informational asymmetries persist?
- How should we trade if we have a monopoly on better information?
- If there are multiple informed traders, will they quickly compete away their advantage?
- In the stock market, private information might come from a few people who have advance knowledge of an earnings announcement or a planned acquisition.
  - But government bond and FX markets also seem to exhibit order price impact.
  - What is "private information" in the FX or government bond market?

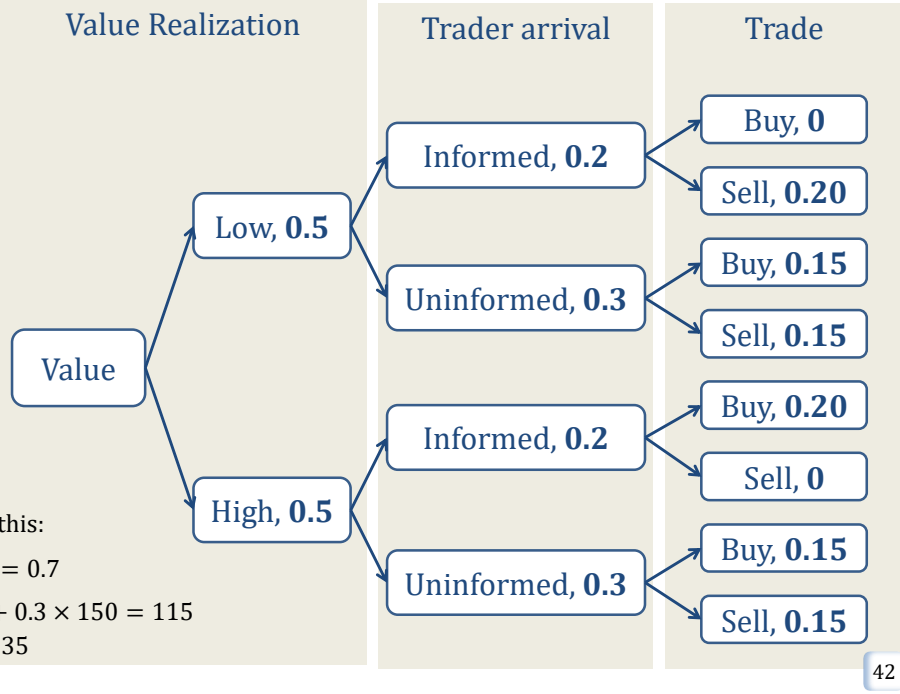
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Answer to embedded problem 1.

- On the ask side. Given that there's a *buy* …
- $P(V = Low|Buy) = \frac{P(V=Low,\ Buy)}{P(Buy)} = \frac{0.2}{0.5} = 0.4$
- The expected security value is
  - $E[V|Buy] = P(V = Low|Buy) \times V_{Low} + P(V = High|Buy) \times V_{High}$
    $= 0.4 \times 100 + 0.6 \times 150 = 130$

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Answer to embedded problem 2.



- The tree now looks like this:
- $P(V = Low|Sell) = \frac{0.35}{0.5} = 0.7$
- $E(V|Sell) = 0.7 \times 100 + 0.3 \times 150 = 115$
- Similarly, $E(V|Buy) = 135$

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Answer to embedded problem 3

- Conditional probabilities

$$P(V = Low|Buy_2) = \frac{P(V = Low, Buy_2)}{P(Buy_2)} = \frac{0.24}{0.08 + 0.24 + 0.16} = 0.5$$

$$P(V = High|Buy_2) = 0.5$$

- A *buy* followed by a *sell* leaves the dealer with the same beliefs as he held initially.
  - When the order flow is balanced we learn nothing.
- Conditional expectations

$$E(V|Buy_2)$$
$$= P(V = Low|Buy_2) \times 100 + P(V = High|Buy_2) \times 150 = 125$$

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